

COURSE DESCRIPTION			
Program	Textile Technology		
Course Title	Fundamentals of Textile Mechanics Lab		
Course Code	2069	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

The Fundamentals of Textile Mechanics Lab provides practical exposure to mechanical elements and power transmission systems used in textile machinery, enabling students to apply theoretical concepts through hands-on experiments and develop essential engineering skills.

Course Objectives

1	Familiarise basic mechanical elements used in textile machines and describe their functions.
2	Understand different types of power transmission systems used in textile machinery.
3	Explain the basic concept of speed calculations in simple drive systems.
4	Demonstrate the working principles of cams, tappets, and variable speed drives.

Course Pre-requisite

Topic	Course code	Course Title	Semester
NIL			

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Familiarise with common mechanical elements used in textile machinery.	Understand
CO2	Explain basic power transmission systems used in textile machines.	Apply
CO3	Apply basic formulas to calculate speed in simple gear and belt drive systems.	Apply
CO4	Understand the working principles of cam, tappet and Variable Speed Drive mechanisms.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	3	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	12	3	-	-	-	-	-	-	-	-	-
Correlation Level	3	3	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Identification of Mechanical Elements in Textile Machinery	Identify and list basic mechanical elements used in textile machines.	CO1	Understand	3
2	Study of Shafts, Axles, Pulleys, Keys and Couplings in Textile Machines	Explain the function of shafts, axles, pulleys, keys and couplings used in textile machines.	CO1	Understand	3
Module II					
3	Study of Flexible and Rigid Drives	Explain the difference between flexible and rigid drive systems.	CO2	Understand	3
4	Study of Belt Drives in Textile Machines	Explain power transmission principles to identify and select suitable belt drives for textile machinery applications	CO2	Understand	3
5	Study of Roller Chain Drives	Describe the working of roller chain drives.	CO2	Understand	3
6	Measurement of RPM and Surface Speed	Apply basic formulas to calculate RPM and surface speed using measured values from the tachometer.	CO2	Apply	4
	Series Exam - I		CO2	Apply	3
Module III					
7	Study of Gears in Textile Machines	Identify different types of gears and state their functions.	CO3	Understand	3
8	Study of Worm, Worm Wheel, Pawl and Ratchet Drives	Describe the working principles of worm and ratchet mechanisms.	CO3	Understand	4
9	Speed Calculations in Gear Train Systems	Study the gear train and drive transmission system, draw the gearing diagram, and calculate the speed and belt slip percentage.	CO3	Apply	3
Module IV					
10	Study of Cam and Tappet Mechanisms	Describe the working of cam and tappet mechanisms.	CO4	Understand	3
11	Study of Variable Speed Drives in Textile Machines	Explain the principle of variable speed drives in textile machines, including stepless cone pulleys and PIV drives.	CO4	Understand	4
	Open-ended experiment		CO4	Apply	3
	Series Exam - II		CO4	Apply	3

Open-ended experiment**Objective:**

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
 Integrate multiple concepts from the syllabus into a combined application.
 Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
 Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare a short note by watching youtube video on list of basic mechanical parts and assemblies	1.5
2	Make a table listing each element, its type, and its primary function.	1.5
3	Watch tutorial videos or animations showing the operation of shafts, pulleys, and axles in textile machines. & submit a note based on the video you have watched.	1.5

Module II		
4	Prepare comparison charts of drive systems.	1.5
5	Summarise which drives are preferred for high-speed versus high-torque applications.	1.5
6	Justify the selection of a specific belt drive for a given machine scenario.	1.5
7	Explain why a flexible or rigid drive, a specific belt drive, or a roller chain drive is suitable for each case.	1.5
Module III		
8	Study different types of gears (spur, helical, bevel, and worm gears) and their characteristics, and submit a table in one page.	1.5
9	Prepare a labelled chart showing each gear type, its function, and the machine it is used in.	1.5
10	Draw a sample drive transmission/gearing diagram from any textile machine.	1.5
11	Study simple and compound gear trains by watching videos, and submit a note based on the video you have watched.	1.5
12	Practice numerical problems on speed calculations.	1.5
Module IV		
13	Review cam and tappet mechanisms by watching videos. Submit a note based on the video you have watched.	1.5
14	Prepare a diagram showing speed variation and explain how speed control is achieved in variable speed drives.	1.5
15	Identify textile machines using PIV drives, and submit a note based on your study.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	T1 R2 - Textile mechanisms for Spinning and Weaving machines	Ganapathy Nagarajan	
2	Mechanics of Spinning machines	Rengasamy R.S	NCUTE

REFERENCE			
Sl. No.	Title	Author	Publication
1	Laboratory manual		
2	Mechanics and calculations of Textile Machinery	R. Gokarneshan,	

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/	All
2	www.sciencedirect.com./machine-design	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Spinning machineries		
2	Power looms		
3	Knitting machines		
4	Processing machines		
5	Tachometer	Non-contact type (2.5-99999 rpm)	2

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to			15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50